

Central Dogma of Biology: genetic structure in plants

Refers to the flow of information necessary for the existence of life.

DNA → RNA → Protein

Replication

Transcription

Translation

Reverse Transcription

What is the difference between a nuclear and a chloroplast genome? Significance

Feature	Nuclear Genome	Chloroplast Genome (cpDNA)
Structure	Linear chromosomes; highly variable size due to polyploidy and transposable elements.	Small, circular (usually), quadripartite structure (LSC, SSC, and two IR regions).
Size	Enormous variability (e.g., 135 Mb to 149 Gb).	Relatively stable and small (typically 115–165 kb).
Inheritance	Biparental.	Uniparental (typically maternal); paternal DNA is usually excluded.
Evolution	Fast; subject to recombination and rapid nucleotide substitution.	Slow; highly conserved coding regions with limited to no recombination.
Gene Count	Tens of thousands of genes (e.g., ~27,000 in <i>Arabidopsis</i>).	~120–130 genes (mostly involved in photosynthesis, transcription, and translation).
Nomenclature	Uppercase (e.g., RBCS).	Lowercase (e.g., rbcL).

The endosymbiotic theory explains the origin of the chloroplast genome. Chloroplasts are believed to have originated when an ancestral cyanobacterium was engulfed by an early eukaryotic cell.¹²³

As these cyanobacteria became established organelles within the host cell, they brought their own genetic material with them. Over time, through a process known as intracellular gene transfer, a significant portion of the original bacterial genetic material was transferred to the host's nuclear genome. The modern chloroplast genome (or plastome) exists today as a remnant of that ancestral cyanobacterial DNA, which remains separated from the nuclear genome within the organelle.²³¹⁴

C3 vs C4

C3 and C4 plants differ primarily in their photosynthetic efficiency and carbon fixation

mechanisms, which are driven by distinct spatial gene expression patterns. Differences in Photosynthesis

Efficiency: C3 plants use an ancient, less efficient method for carbon fixation compared to C4 plants. C4 plants have evolved to more effectively assimilate raw materials and solar energy.

Spatial Separation: C4 photosynthesis is characterized by the spatial separation of carbon fixation, which is not present in C3 plants.

Gene Expression Patterns in C4 Plants

The shift to C4 photosynthesis relies on significant differences in gene expression:

Spatial Differentiation: C4 plants exhibit specialized expression where mesophyll cells express high levels of PEP carboxylase (PEPC), while bundle sheath cells preferentially express Rubisco.

Regulatory Mechanisms: The expression of C4-related genes is controlled at trans-acting, cell-specific, and individual gene levels. Regulation involves the presence or absence of specific activators and repressors.

Transformation: Efforts to convert C3 crops (like rice) into C4-type plants involve introducing C4 pathway-specific genes (such as PEPC, NADP-MDH, and NADP-ME) and C4-related transporter genes.

1. Which genomic feature is most characteristic of plant nuclear genomes compared with animal nuclear genomes?

- A. Absence of repetitive DNA
- B. Extensive gene duplication resulting from recurrent whole-genome duplication events
- C. Lack of transposable elements
- D. Smaller genome sizes and lower chromosome numbers

Answer: B

Explanation: Plants frequently undergo polyploidization and whole-genome duplication (WGD), generating large gene families and functional redundancy. Animal genomes generally experience fewer WGD events.

2. A researcher discovers a gene encoding a photosystem II core protein. The gene is located in the chloroplast genome. Which protein is most likely encoded by this gene?

- A. RBCS
- B. CAB/LHCB
- C. PsbA
- D. HY5

Answer: C

Explanation: PsbA encodes the D1 protein of Photosystem II and is typically chloroplast-encoded. RBCS, LHCB, and HY5 are encoded in the nuclear genome.

3. Which observation provides the strongest evidence supporting the endosymbiotic origin of plastids?

- A. Plastids contain chlorophyll pigments.
- B. Plastids are enclosed by lipid membranes.
- C. Plastids possess circular DNA and bacterial-type ribosomes.
- D. Plastids participate in photosynthesis.

Answer: C

Explanation: Circular genomes and 70S bacterial-like ribosomes are hallmark features supporting the hypothesis that plastids originated from a cyanobacterial ancestor through endosymbiosis.

4. During evolution, many ancestral plastid genes were transferred to the host nucleus. Which process best explains this phenomenon?

- A. Horizontal gene transfer from fungi
- B. Endosymbiotic gene transfer (EGT)
- C. Meiotic recombination
- D. RNA editing

Answer: B

Explanation: Endosymbiotic gene transfer refers to the movement of genes from the endosymbiont (ancestral cyanobacterium/plastid) into the host nuclear genome, a major event in plastid evolution.

5. A mutation eliminates expression of the nuclear gene encoding the chloroplast-targeting transit peptide of a photosystem protein. What is the most likely consequence?

- A. The protein will accumulate in the nucleus.
- B. The protein will be correctly imported into chloroplasts.
- C. The protein will fail to reach the chloroplast despite being synthesized.
- D. Chloroplast DNA replication will increase.

Answer: C

Explanation: Most chloroplast proteins are encoded in the nucleus, translated in the cytoplasm, and imported into chloroplasts via N-terminal transit peptides. Loss of the targeting sequence prevents chloroplast import.

6. Which gene expression pattern is most characteristic of C4 photosynthesis?

- A. Rubisco and PEP carboxylase expressed equally in all photosynthetic cells
- B. PEP carboxylase enriched in mesophyll cells; Rubisco enriched in bundle sheath cells
- C. Rubisco absent from bundle sheath cells
- D. PEP carboxylase exclusively expressed in chloroplasts

Answer: B

Explanation: C4 plants exhibit spatial separation of carbon fixation. Mesophyll cells express high levels of PEP carboxylase (PEPC), while bundle sheath cells preferentially express Rubisco.

7. Which gene would be expected to show significantly higher expression in C4 mesophyll cells compared with C3 mesophyll cells?

- A. PEPC
- B. RBCS
- C. rbcL
- D. PsbA

Answer: A

Explanation: PEPC is the primary carbon-fixing enzyme in C4 mesophyll cells, converting bicarbonate into oxaloacetate. Its expression is dramatically elevated in C4 plants.

8. The chloroplast genome of most angiosperms is approximately 120–170 kb in size and exhibits which structural organization?

- A. Linear chromosome with telomeres
- B. Single circular chromosome lacking repeats
- C. Circular genome containing large and small single-copy regions separated by inverted repeats
- D. Multiple chromosomes with centromeres

Answer: C

Explanation: Most land plant plastomes display a quadripartite structure consisting of a Large Single Copy (LSC), Small Single Copy (SSC), and two Inverted Repeat (IR) regions.

9. A scientist compares expression of photosynthesis genes during a transition from darkness to light. Which transcription factor is most directly involved in activating nuclear-encoded photosynthetic genes in response to light?

- A. HY5
- B. DCL1
- C. AGO1
- D. SPO11

Answer: A

Explanation: HY5 is a central regulator of photomorphogenesis and activates numerous nuclear-encoded photosynthesis genes upon light exposure.

10. Which evolutionary scenario best explains why chloroplast genomes encode only a small fraction of the proteins required for photosynthesis?

- A. Chloroplast genomes evolved independently of cyanobacteria.
- B. Most photosynthesis genes originated in the nucleus.
- C. Extensive endosymbiotic gene transfer moved many ancestral plastid genes to the nucleus.
- D. Chloroplasts acquire proteins directly from mitochondria.

Answer: C

Explanation: Modern chloroplasts retain only about 80–120 genes, while thousands of chloroplast proteins are encoded by nuclear genes. This is largely the result of endosymbiotic gene transfer during evolution.